

Chapter 1 Start developing App with MIT App Inventor 2

Key Words

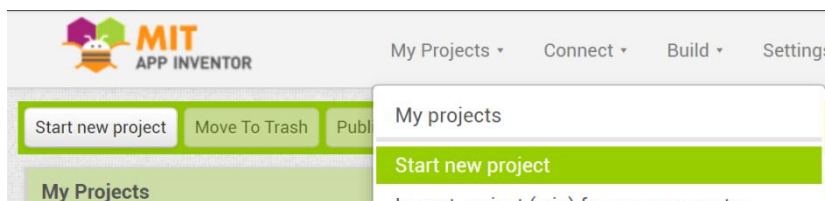
Online Platform 網上平台	Component 元件 / App 中的物件	Visible 可視的
Non-visible 不可視的	Interface 界面 / 程式畫面	

1.1 Create a project in online platform¹: App Inventor 2

- Step 1: Open URL: <http://ai2.appinventor.mit.edu/> in Google Chrome.



- Step 2: Login with Google (Use your school Google account).
- Step 3: Add a new project:
Select "My Projects" → "Start new project"
 or
Click the button "Start new project"



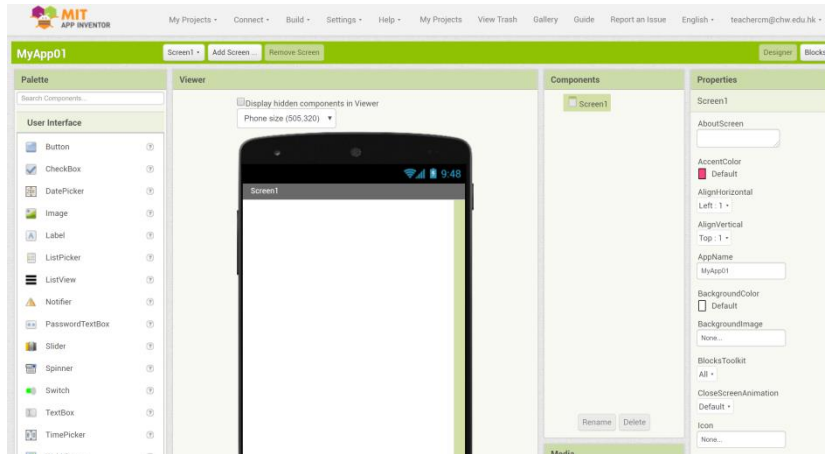
*** You should enter a name,

- begin with a letter,
- contain no spaces,
- only contain letters, numbers and underscores.

¹ Online Platform 網上平台

Notes

- Step 4: You may now start developing your first App.



Notes

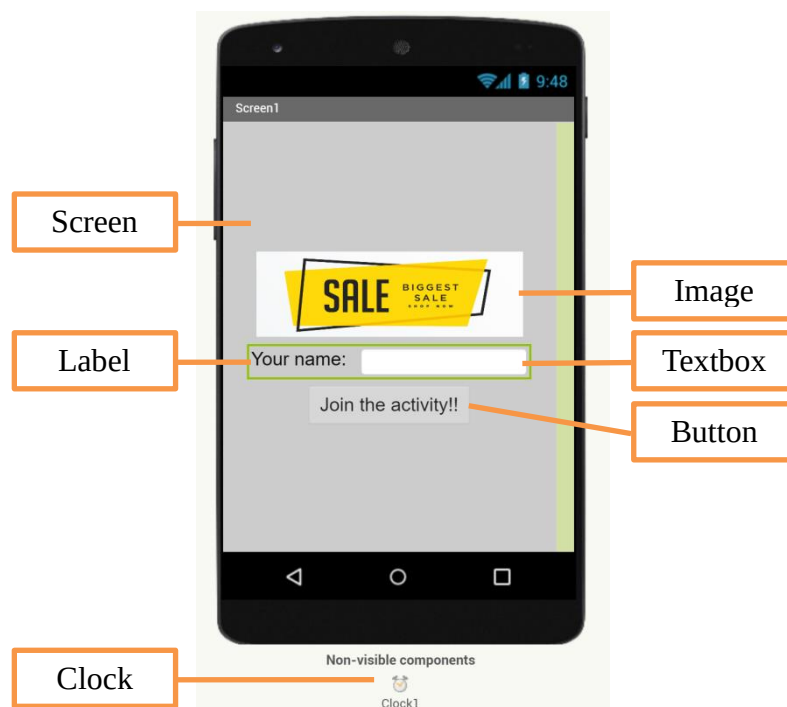
1.2 Introduction to five common components¹

Examples of visible² component:

- User Interface → Buttons – It is used to detect click.
- User Interface → Image – It is used to display image/picture.
- User Interface → Label – It is used to show text.
- User Interface → TextBox – It is used for user text input.

Example of non-visible³ component:

- Sensors → Clock – It is used to fire a timer to perform specific task.



Example of an user interface

¹ Component 元件 / App 中的物件

² Visible 可視的

³ Non-visible 不可視的

1.3 Homework

Task 1:

- Finish "Worksheet 01". Name the component in the **interfaces**¹ given.
- Scan your homework into PDF format and submit to the Schoology.

Mobile App Development Name: _____ () Class: _____

Worksheet 01 – Interface design with different components

We should use the following components to build the App. Finish Task 1 with the following names.

1. Screen	2. Button	3. Label	4. Image	5. Clock
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Task 1 (20 marks): Write down the component names in the boxes below.

Task 2 (5 marks): Create a project with *App Inventor 2* and upload the exported project file (.aia) to the Schoology: (<http://ai2.appinventor.mit.edu/>)

Worksheet 01

Task 2:

- Login MIT App Inventor 2 (<http://ai2.appinventor.mit.edu/>).
- Start a new project.
- Add some components on Screen1. E.g. Image, Label, Textbox, Button, ...
- Export the project into (.aia) format and submit to the Schoology.

¹ Interface 界面 / 程式畫面

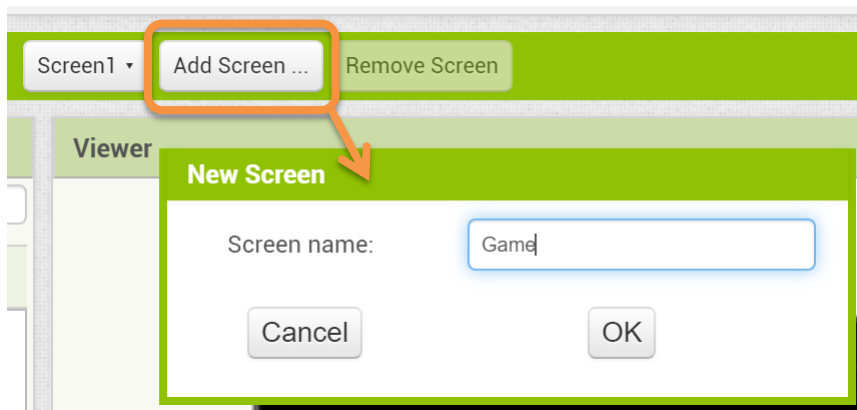
Chapter 2 Steps of designing interface of an App

Key Words

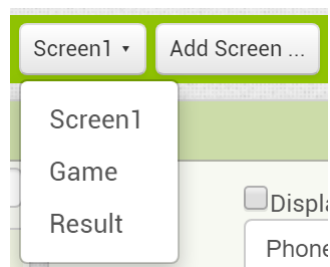
Default 預設	Layout 佈局 / 配置 / 設計	Property / Properties 屬性
Scrollable 可捲動	Grid 格 / 格子	

2.1 First step of developing App: Add Screen

- By default¹, the screen "Screen1" should be the startup screen.
- You should design the structure of your App by adding the screens needed.
- For example, adding a screen "Game".



- After adding two screens, "Game" and "Result", the following figure shows an App with 3 screens.

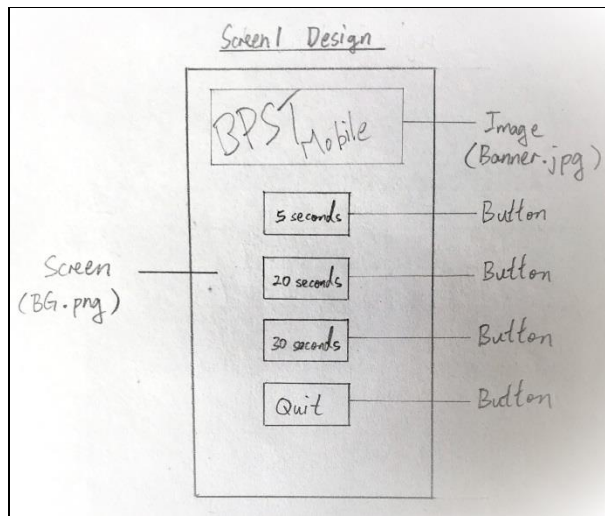


Notes

¹ Default 預設

2.2 Second step of developing App: Interface design

- You should design the interface of every screen.



An example of the Main Menu interface design

- When designing an interface, you should consider the followings:
 1. **Layout**¹ of the interface;
 2. Components needed on the screen;
 3. **Properties**² of every components.

Layout – The way in which text or pictures are set out on a screen.

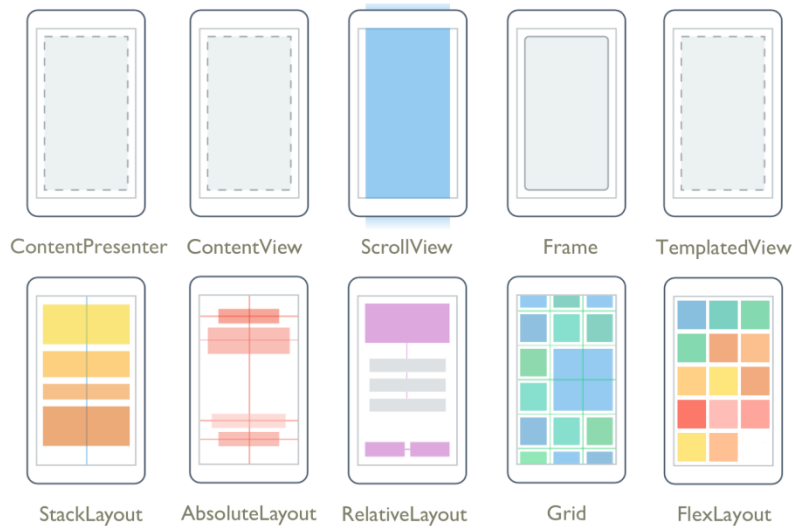


Image source:

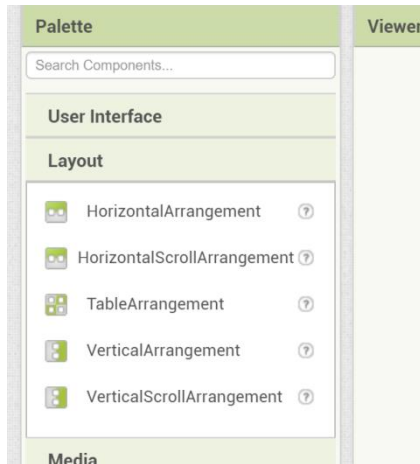
<https://docs.microsoft.com/zh-tw/xamarin/xamarin-forms/user-interface/controls/layouts>

¹ Layout 佈局 / 配置 / 設計

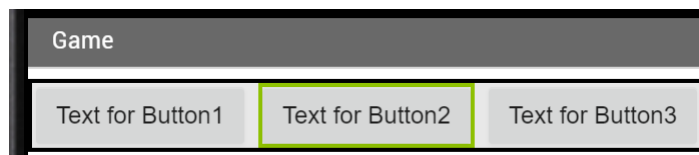
² Property / Properties 屬性

- In App Inventor 2, there are several types of layout component.

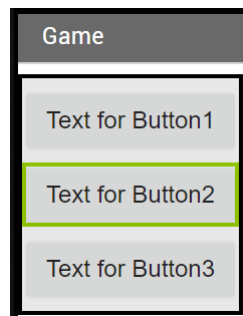
Notes



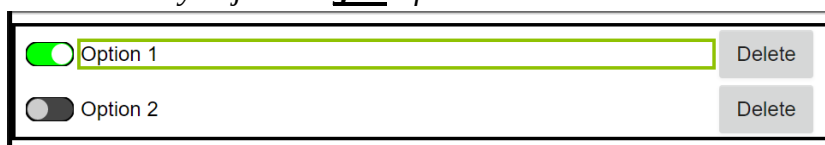
1. *Layout* → *HorizontalArrangement* or
Layout → *HorizontalScrollArrangement*
 - Arrange objects from left to right
(non-scrollable or scrollable¹)



2. *Layout* → *VerticalArrangement* or
Layout → *VerticalScrollArrangement*
 - Arrange objects from top to bottom
(non-scrollable or scrollable)



3. *Layout* → *TableArrangement*
- Array objects in grid² form



¹ Scrollable 可捲動

² Grid 格 / 格子

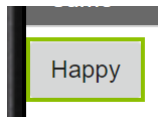
Components – the objects in an App.(Reference: <http://ai2.appinventor.mit.edu/reference/components/>)

- The following table shows some components in the categories "User Interface" and "Layout". (You can refer the reference site above to see the full list of components and their functions)

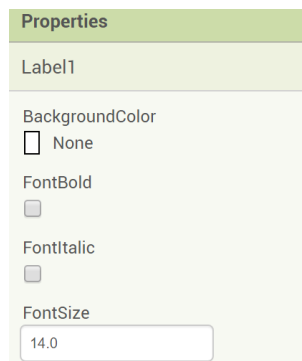
<u>User Interface</u>	<u>Layout</u>
Button	HorizontalArrangement
CheckBox	HorizontalScrollArrangement
Image	TableArrangement
Label	VerticalArrangement
PasswordTextBox	VerticalScrollArrangement
Screen	
Switch	
TextBox	

Notes**Properties** – The values of the properties represent the state of the component.

- E.g. Button.Text = "Happy"
It means that the text "Happy" shows on the button.



- E.g. Label1.BackgroundColor = Red
It means that set the background color of Label1 to red.
- E.g. Label1.FontSize = 20
It means that set the font size of text on Label1 to point 20.

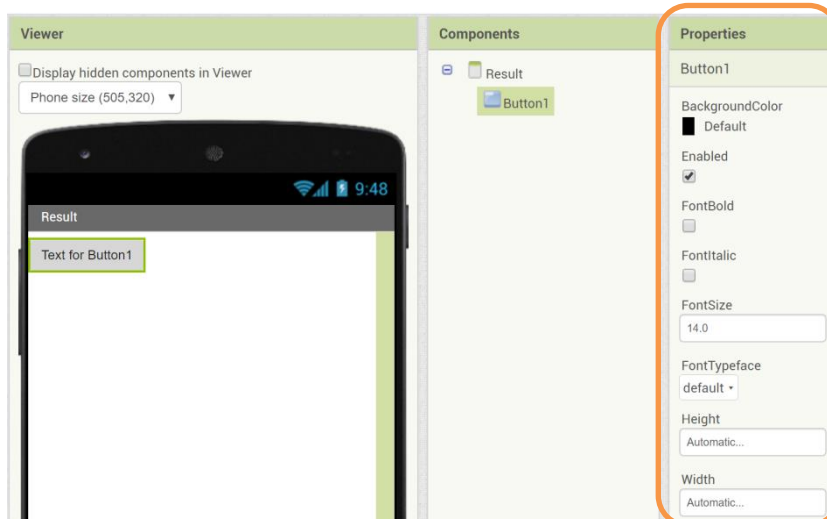


Add component to the Screen and change its properties.

- Add components into every screen by dragging components into screens.



- Select the component and change its properties by changing the values in the properties window (at the right-most area).



- The following is the introduction of some common properties.
 1. *BackgroundColor* – It indicates the color of the component.
 2. *Enabled* – It indicates user can use this component or not.
 3. *FontSize* – It indicates the size of text on the component.
 4. *Height* – It indicates the height of the component.
 5. *Width* – It indicates the width of the component.
 6. *Text* – It indicates the text shown on the component.
 7. *Visible* – It indicates the component can be seen or not.
- You can find all properties introduction of components in the web-site (<http://ai2.appinventor.mit.edu/reference/components/>).

Notes

2.3 Homework

Task 1 (10 marks):

You are a mobile App developer. Before program the App "Button Pressing Speedtest (BPST Mobile)" in App Inventor 2, you should design 3 interfaces (Screen1, Game, Result) by

1. Sketching the layout and components on each screen.
2. Name the components on each screen.

You should design the interfaces "Game" and "Result" by drawing.

Task 2 (5 marks):

In MIT App Inventor 2, finish the following steps:

Step 1: Start a new project named "BPST_3E01". (Please use your own class and class number)

Step 2: Use "Add Screen..." function to add 2 screens named "Game" and "Result".

Step 3: In every screen, try adding components according to your design in Task 1 and design the interface "Screen1" as well.

Step 4: Try to edit the "Properties" of every components.

Step 5: Export your project to (.aia), and submit to the Schoology.